

Script-Controlled ACL – Restrict Record Access Based on Field Value

Project Demonstration Phase

Description

This lab demonstrates how a script-controlled Access Control List (ACL) restricts users from viewing records unless they meet specific conditions. Only users belonging to the EEE branch can view records where the Branch field is set to EEE, while administrators retain full access.

Timer: 3 hrs

Demonstration Setup

1. Impersonate the EEE user (role bb1) and open the Student Records (Institution Details) list.
2. Copy the table name u_institution_details.list/LIST and paste it in the All menu to open the list directly.
3. Impersonate a user with no assigned role and open the same list.
4. Impersonate Admin and open the Student Records list.
5. Repeat impersonation for users holding bb1+bb2, bb1+bb2+bb3, and bb1+bb2+bb3+bb4 to demonstrate Create, Write, and Delete access respectively.

Demonstration of Read ACL

The Read ACL is scripted to allow access only when the user has the admin role or the custom bb1 role, combined with a data condition that the Branch field equals EEE:

```
(function () {  
  // Allow admin users full access  
  if (gs.hasRole('admin')) {  
    return true;  
  }  
  // Allow only EEE branch users to see EEE records  
  if (gs.hasRole('bb1')) {  
    return true;  
  }  
  // Deny access for all others  
  return false;  
})();
```

Verify

- User with the bb1 role can view only EEE branch records.
- User without the bb1 role cannot view any records.
- Admin user can view all records regardless of branch.

Demonstration of Create ACL

The Create ACL requires the custom bb2 role, with no additional data condition, so any user holding bb2 can create new Institution Details records.

Verify

- User with bb1 & bb2 roles can view only EEE branch records, and the New button is now available to create records.

Demonstration of Write ACL

The Write ACL requires the custom bb3 role, with no additional data condition, allowing users with this role to edit existing records.

Verify

- User with bb1, bb2 & bb3 roles can view, create, and edit the EEE branch records (e.g. updating the Description field, which saves successfully).

Demonstration of Delete ACL

The Delete ACL requires the custom bb4 role, with no additional data condition, allowing users with this role to delete records.

Verify

- Users with bb1, bb2, bb3 & bb4 roles can see, create, edit, and delete the EEE branch records.
- Selecting a record and choosing Delete from the Actions on selected rows menu, followed by confirming in the dialog, successfully removes the record.

Outcome

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.